

Disproof of Conjecture 00000002141: there are two different cospectral pairs of 4-regular graphs of order 10

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Abstract

Conjecture 00000002141 asserts that the minimal order of a pair of cospectral (non-isomorphic) regular graphs is 10, and that the cospectral regular pair of order 10 is unique. We exhibit two cospectral pairs (A_1, B_1) and (A_2, B_2) of 4-regular graphs on 10 vertices. The two pairs have different spectra, and no graph of the first pair is isomorphic to a graph of the second. So the uniqueness clause is false. An exhaustive enumeration shows that order 10 has exactly two 4-regular cospectral pairs and their two 5-regular complements. The disproof is formalized in Lean 4 (core library only).

1 The conjecture

Definition: Cospectral regular graphs are non-isomorphic regular graphs with the same adjacency spectrum. *Conjecture:* The minimal order of a cospectral regular pair is 10 (the Saltire pair); the cospectral regular pair of order 10 is unique (classification uniqueness).

The Chinese version says the same (the order-10 pair is the only one). The conjecture is a conjunction, so it suffices to refute the uniqueness clause. We read “the pair is unique” in the weakest sense: any two cospectral regular pairs $\{G, H\}$ and $\{G', H'\}$ of order 10 coincide up to isomorphism as unordered pairs. (The first clause is in fact true; see Section 5. The name “Saltire pair” usually refers to the non-regular pair $K_{1,4}, C_4 \cup K_1$ on 5 vertices, but this plays no role.)

2 The graphs

All four graphs have vertex set $\{0, \dots, 9\}$ and 20 edges.

A_1 : 03, 05, 07, 09, 14, 15, 17, 19, 25, 26, 27, 28, 36, 38, 39, 46, 48, 49, 57, 68

B_1 : 03, 05, 07, 08, 14, 15, 18, 19, 25, 26, 27, 29, 36, 37, 38, 46, 48, 49, 57, 69

A_2 : 03, 05, 07, 08, 14, 15, 17, 19, 25, 26, 28, 29, 36, 37, 38, 46, 48, 49, 57, 69

B_2 : 03, 04, 06, 07, 15, 16, 18, 19, 25, 27, 28, 29, 34, 36, 37, 48, 49, 57, 58, 69

(In graph6 format: ICR‘vGyu?, ICR‘tjWY_, ICR‘uiwY_, ICdedhkY_.) Each vertex has degree 4 in each graph.

3 Cospectrality

For a graph with adjacency matrix A and eigenvalues $\lambda_1, \dots, \lambda_n$, the number of closed walks of length k is $\text{tr}(A^k) = \sum_i \lambda_i^k = p_k$. By Newton's identities $ke_k = \sum_{i=1}^k (-1)^{i-1} e_{k-i} p_i$, the power sums $p_1, \dots, p_n$ determine the elementary symmetric functions $e_1, \dots, e_n$, hence the characteristic polynomial $\prod (x - \lambda_i)$, hence the spectrum. The converse is clear. So two graphs on n vertices are cospectral if and only if they have equally many closed walks of each length $k \leq n$.

Proposition 1. $\text{tr}(A_1^k) = \text{tr}(B_1^k)$ and $\text{tr}(A_2^k) = \text{tr}(B_2^k)$ for $0 \leq k \leq 10$. Equivalently,

$$\begin{aligned} \det(xI - A_1) &= \det(xI - B_1) = x^{10} - 20x^8 - 16x^7 + 110x^6 + 136x^5 \\ &\quad - 180x^4 - 320x^3 + 9x^2 + 200x + 80, \\ \det(xI - A_2) &= \det(xI - B_2) = x^{10} - 20x^8 - 14x^7 + 108x^6 + 104x^5 \\ &\quad - 183x^4 - 188x^3 + 80x^2 + 68x - 16. \end{aligned}$$

The two polynomials differ. For example, $\text{tr}(A^3)$ is six times the number of triangles, and $\text{tr}(A_1^3) = 48$ (8 triangles) while $\text{tr}(A_2^3) = 42$ (7 triangles).

Proof. Direct computation, in Lean (`cospectral_1`, `cospectral_2`, `spectra_differ`) and in `verify.py`. The script computes the characteristic polynomials in two independent exact ways: Faddeev–LeVerrier, and $\det(xI - A)$ by exact Gaussian elimination at $x = 0, \dots, 10$. $\square$

4 Non-isomorphism

Call an edge uv *triangle-free* if u and v have no common neighbour. Both of the following properties are preserved by graph isomorphisms:

- (P) some edge is triangle-free;
- (Q) some triangle-free edge shares no endpoint with any other triangle-free edge.

Lemma 2. A_1 fails (P), while B_1 , A_2 and B_2 satisfy (P). A_2 satisfies (Q), while B_2 fails (Q).

Proof. The triangle-free edges are: none in A_1 ; 15, 36 in B_1 ; 25, 28, 36, 48 in A_2 , where 36 is isolated; and 29, 48, 49 in B_2 , which form the path 2–9–4–8. $\square$

Hence $A_1 \not\cong B_1$ and $A_2 \not\cong B_2$, so both are cospectral regular pairs. Also $A_1 \not\cong A_2$ and $A_1 \not\cong B_2$, so the unordered pairs $\{A_1, B_1\}$ and $\{A_2, B_2\}$ are different up to isomorphism. The two pairs also have different spectra.

Theorem 3. The cospectral regular pair of order 10 is not unique. Conjecture 00000002141 is false.

Remark 4. The two pairs are not complements of each other: all four graphs are 4-regular, while complements of 4-regular graphs on 10 vertices are 5-regular. So uniqueness fails even up to complementation.

5 Enumeration

With nauty’s `geng`, `verify.py` enumerates all k -regular graphs on $n \leq 10$ vertices and groups them by characteristic polynomial. There is no cospectral pair of regular graphs on fewer than 10 vertices. On 10 vertices there are exactly two cospectral pairs of 4-regular graphs, namely the two above (out of 60 4-regular graphs), and exactly two of 5-regular graphs, namely their complements. So the first clause (minimal order 10) is true, and there are exactly four cospectral regular pairs of order 10.

6 Lean formalization

The directory `lean4/` is a self-contained Lean 4.19.0 project (core library only, no Mathlib).

- Graphs on $\{0, \dots, 9\}$ are Boolean adjacency functions. `IsSimple` means symmetric and loop-less, and `Regular g k` means every vertex has degree k .
- `closedWalks g k` is $\text{tr}(A^k)$, computed with explicit matrix products. `Cospectral g h` means equal closed-walk counts for all $k \leq 10$, which by the argument above is equivalent to having the same adjacency spectrum.
- `Iso g h` is the existence of a bijection σ of $\{0, \dots, 9\}$ (with inverse τ) such that $g(u, v) = h(\sigma u, \sigma v)$. `iso_hasNT` and `iso_hasIsolatedNT` prove that (P) and (Q) are isomorphism invariants. `invariants` evaluates (P) and (Q) on the four graphs.
- `CospectralRegularPair g h` combines simple, regular, cospectral and non-isomorphic. `UniquePair` says any two such pairs coincide up to isomorphism, in either order. `conjecture_00000002141_false` proves $\neg \text{UniquePair}$, using `pair_1`, `pair_2`, `not_iso_A1_A2` and `not_iso_A1_B2`.

`lake build` succeeds. There is no `sorry`, no `native_decide`, and no added axioms. `#print axioms` shows at most `propext` and `Quot.sound`.

7 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, core only
python3 verify.py          # stdlib (+ nauty-geng if present)
pdflatex report.tex
```

References

- [1] A. E. Brouwer and W. H. Haemers, *Spectra of Graphs*, Springer, 2012 (cospectrality and closed walks; GM switching).
- [2] W. H. Haemers and E. Spence, Enumeration of cospectral graphs, *European J. Combin.* 25 (2004), 199–211.